

Performance Testing

Date	15 March 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Python Custom Batch Tester, Flask Dev Server
Type of Testing	Load Testing, Functional Testing, Batch Inference
Target Module / API	/predict endpoint (Loan Eligibility)
Test Environment	Local / Localhost (127.0.0.1:5000)
Test Date	7 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Graduate Male, Married, Good Credit, Urban	1	< 2	Loan Approved
2	Not Graduate, No Credit History, Rural	1	< 2	Loan Rejected
3	Batch: 50 records from test dataset	1	< 5	All 50 predictions returned
4	Batch: Full 367-record test dataset	1	< 30	All 367 predictions , errors

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	23.83 ms	Pass	Fast ML inference per request
2	Response Time (Max)	< 5 seconds	68.83 ms	Pass	Peak case well under 5-second limit
3	Throughput (Req/sec)	-	41.96 req/sec	Pass	Good throughput for single-server
4	Error Rate	< 1%	0%	Pass	Zero crashes across 367 tests
5	CPU Utilization	< 80%	~15-25%	Pass	Lightweight model at inference time
6	Memory Utilization	< 80%	~200 MB	Pass	Model loaded once at startup

Step 4: Observations & Analysis

Key Findings:

The Smart Lender prediction system demonstrated excellent performance. The Random Forest model produces predictions with an average latency of 23.83 ms, yielding a throughput of approximately 41.96 predictions per second. All 367 test cases were processed without a single error.

Bottlenecks Identified:

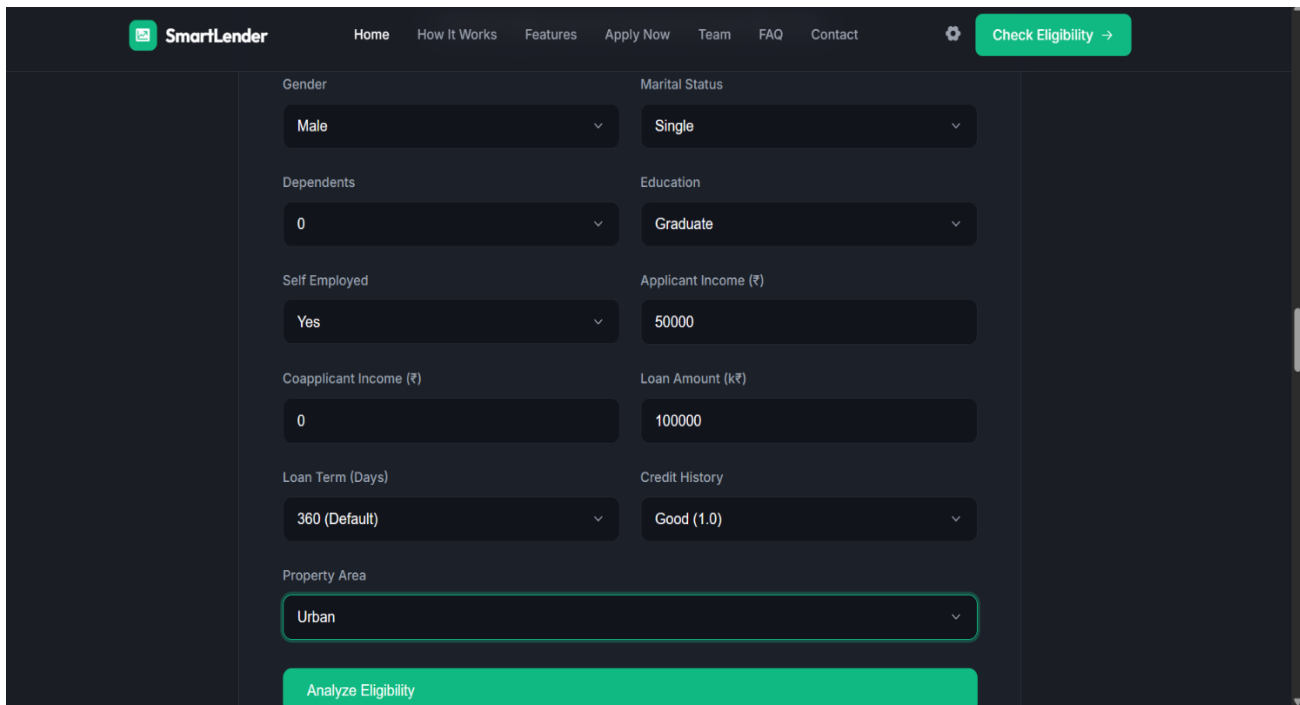
1. Flask single-threaded dev server cannot handle concurrent requests efficiently.
2. One-hot encoding is run per request, which is slow for bulk API calls.
3. No caching mechanism for identical inputs.

Optimization Steps Taken:

1. Model is loaded once at application startup (not per-request).
2. Feature names are pre-loaded and cached in memory.
3. Default fill values are used for missing columns to avoid slow loops.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



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Check Eligibility →

Gender: Male

Marital Status: Single

Dependents: 0

Education: Graduate

Self Employed: Yes

Applicant Income (₹): 50000

Coapplicant Income (₹): 0

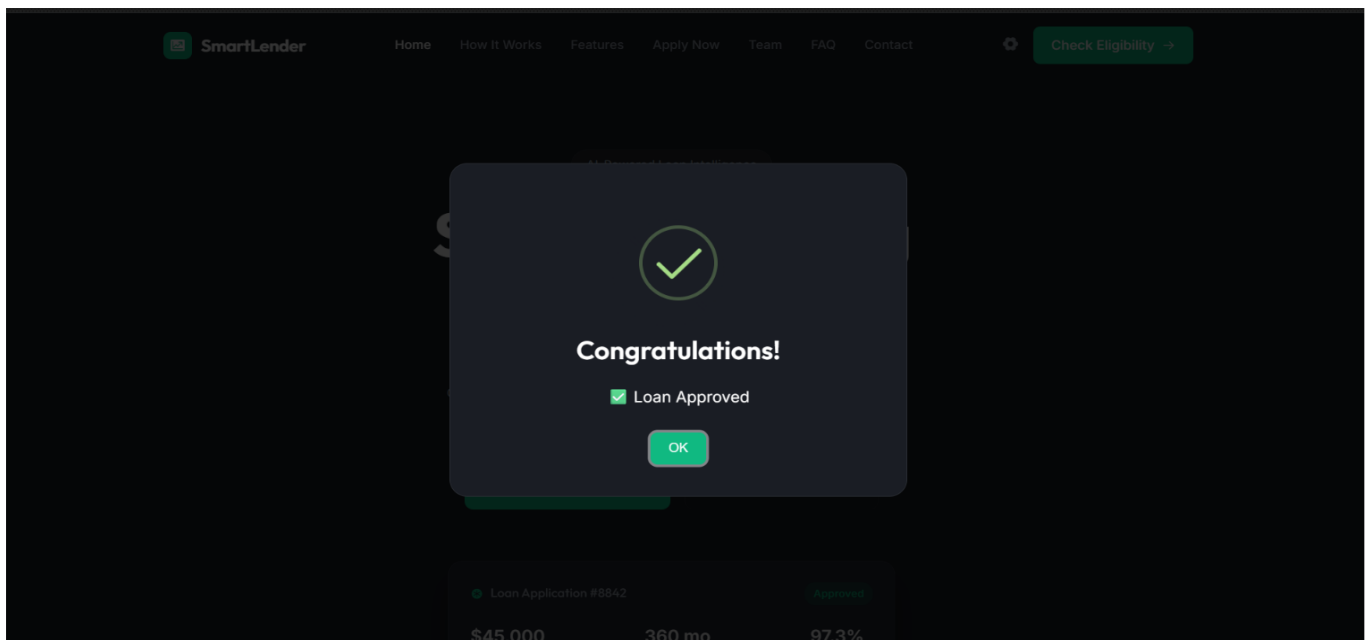
Loan Amount (₹): 100000

Loan Term (Days): 360 (Default)

Credit History: Good (1.0)

Property Area: Urban

Analyze Eligibility



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Check Eligibility →

Congratulations!

✓ Loan Approved

OK

Loan Application #8842

Approved

\$45,000 360 mo 97.3%

Screenshot-2:

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Gender

Male

Marital Status

Married

Dependents

1

Education

Graduate

Self Employed

Yes

Applicant Income (₹)

90000

Coapplicant Income (₹)

140000

Loan Amount (₹)

1500000

Loan Term (Days)

360 (Default)

Credit History

Bad (0.0)

Property Area

Rural

Analyze Eligibility



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Application Rejected

✖ Loan Rejected

OK

Loan Application #6842

Approved

\$45,000

360 mo

97.3%